Containers & OpenShift Workshop

Alexander Mamani - IBM
Kate Queiroz - Red Hat
Christian Zaccaria - Red Hat
Winnie Imafidon - IBM

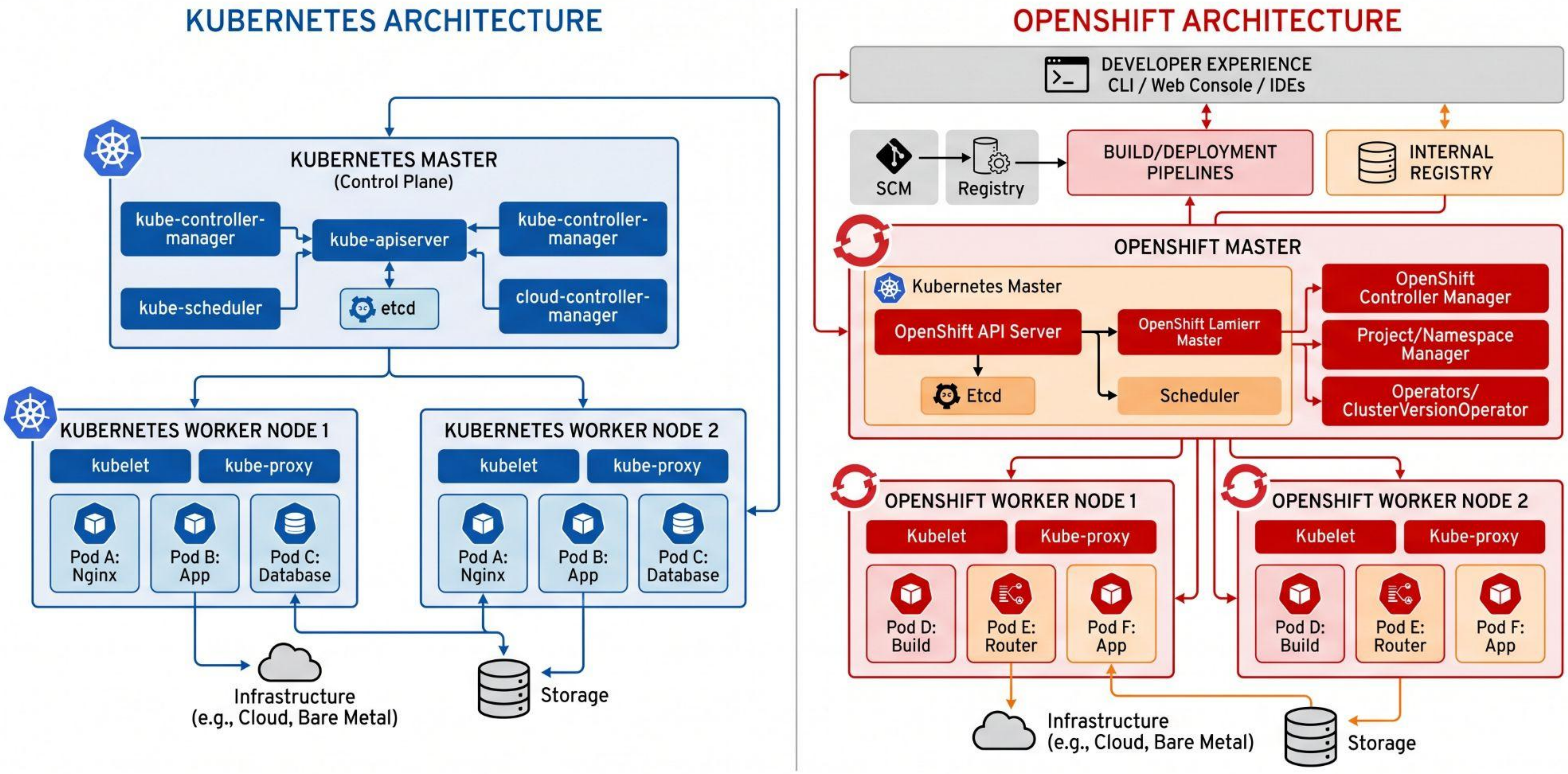
containers are created from container images

Let’s do a recap of kubernetes

1. Kubernetes Recap: What We Covered Yesterday

- Architecture
- Core Objects
- Workloads
- Networking
- Storage
- Configuration

COMPARING KUBERNETES AND OPENSHIFT ARCHITECTURES



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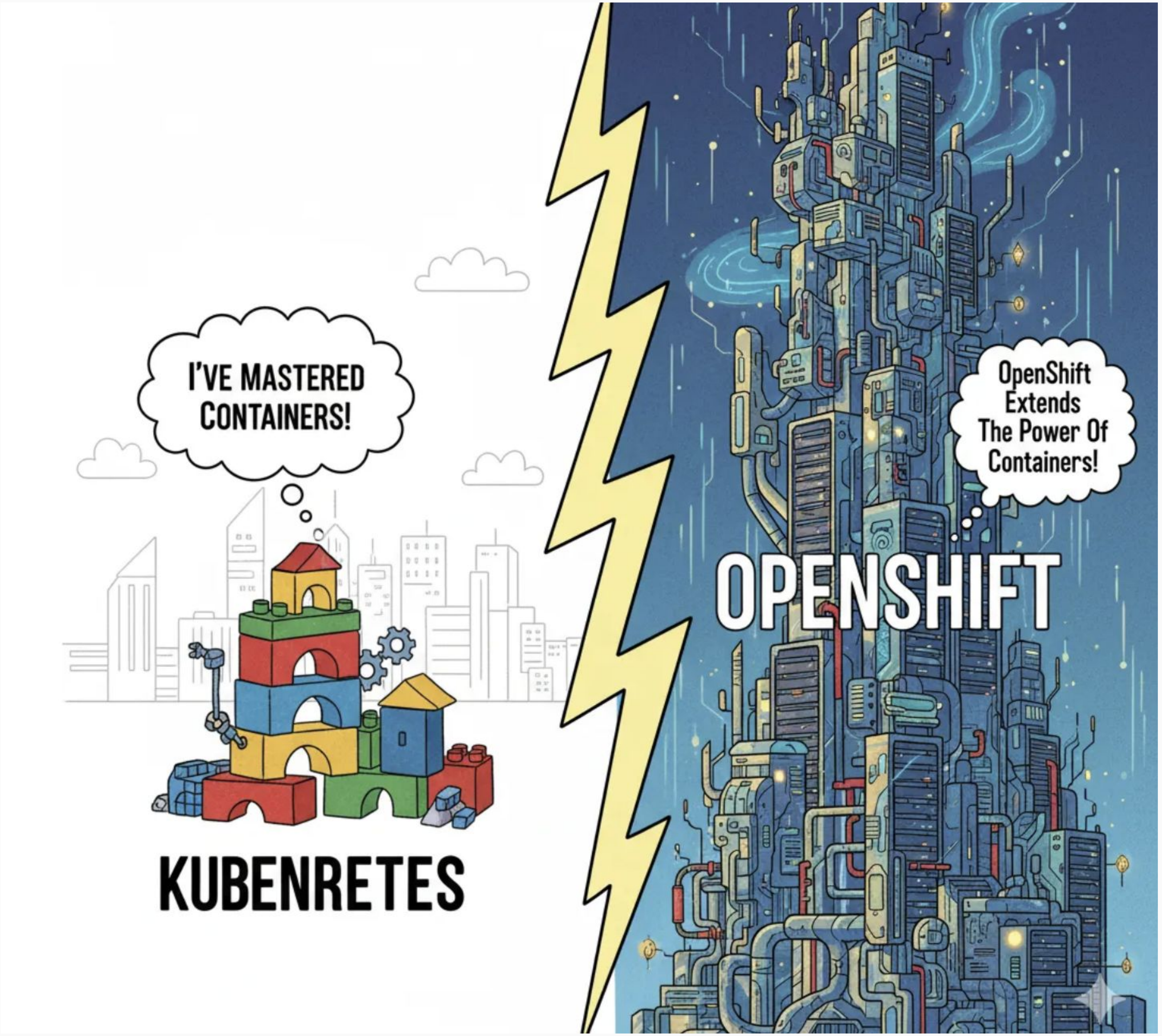
I'VE MASTERED CONTAINERS!
KUBENRETES
OpenShift Extends The Power Of Containers!
OPENSHIFT

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Kubernetes vs OpenShift: Core Platform (Shared Features)

MULTI-CONTAINER, MULTI-HOST SCHEDULING	schedule and orchestrate containers across multiple nodes using the Kubernetes scheduler. They manage workloads, resource allocation, and failover automatically
SELF-SERVICE PROVISIONING	Users can deploy and manage their own applications via declarative YAMLs, CLI, or APIs without needing admin intervention.
SERVICE DISCOVERY	Both use Kubernetes Services and DNS to enable communication between pods via stable name
PERSISTENT STORAGE	Kubernetes Persistent Volumes (PVs) and Persistent Volume Claims (PVCs) allow stateful apps to store data persistently. OpenShift uses the same model.

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Where OpenShift Extends Kubernetes: Enterprise & Security Features

MULTI-TENANCY

NETWORKING (SDN)

IMAGE REGISTRY

IMAGE BUILD TOOLS

METRICS

LOG AGGREGATION

INGRESS

SECURED BY DEFAULT

UX: CONSOLE, SERVICE CATALOG, “OC”

Slide 6 — Side-by-Side: Where OpenShift Extends Kubernetes

Capability	Kubernetes (DIY)	OpenShift (Built-in)
Installation	Manual (kubeadm, kops, k3s)	Automated installer (IPI/UPI)
Host OS	Any Linux distro	RHEL CoreOS (immutable, auto-updated)
Web Console	Dashboard (basic)	Full admin + developer console
Container Registry	External (DockerHub, ECR...)	Integrated registry + ImageStreams
CI/CD	Install Jenkins/ArgoCD yourself	OpenShift Pipelines (Tekton) built-in
Security	PodSecurityAdmission (basic)	SCCs + OAuth + built-in RBAC policies
Networking	Ingress + 3rd-party CNI	Routes (HAProxy) + OpenShift SDN/OVN
Monitoring	Install Prometheus stack yourself	Pre-configured Prometheus + Grafana
Logging	EFK/Loki — manual setup	OpenShift Logging (Loki/Elasticsearch)
Operator Management	OLM optional	OperatorHub + Operator Lifecycle Manager
Developer Experience	kubectl + YAML	oc CLI + oc new-app + S2I + Catalog

Why It Matters for DevOps & Developers

Mastering OpenShift means you're not just able to deploy containers — you're ready to handle real-world enterprise workloads with security, observability, and developer efficiency baked in. Whether it's scaling apps across multiple teams, integrating CI/CD pipelines, or maintaining robust monitoring and logging, OpenShift streamlines the process that Kubernetes alone would leave you wiring together manually.

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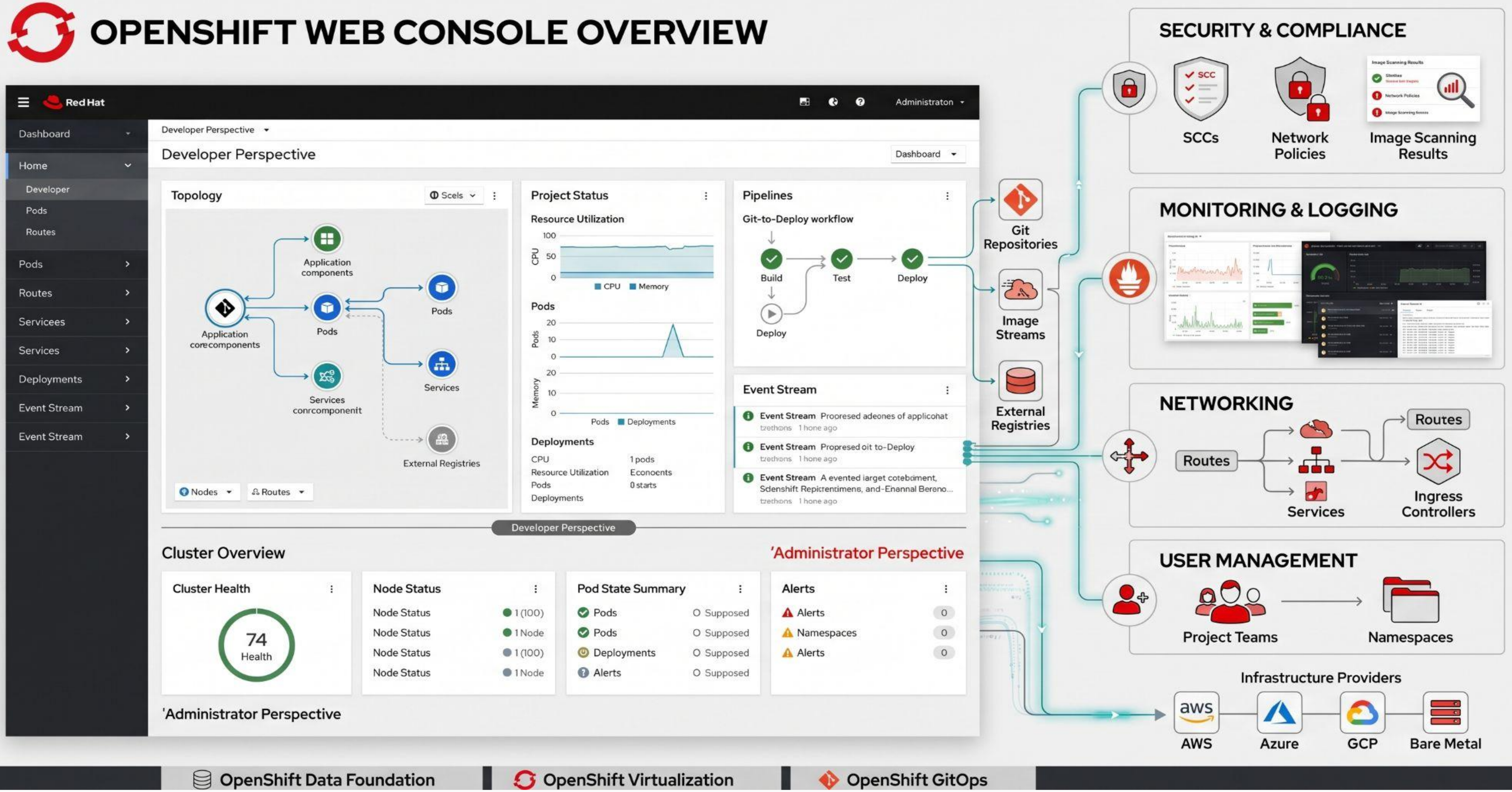
What is OpenShift?

Red Hat OpenShift is a commercial enterprise hybrid-cloud platform (PaaS) built on top of Kubernetes that automates container lifecycle management across any infrastructure. While plain Kubernetes provides the core container orchestration engine, OpenShift acts as a fully packaged, turnkey system by integrating built-in developer tools—like automated Source-to-Image (S2I) build pipelines, a local container registry, and a web console—alongside production-ready enterprise defaults including rigid security constraints (SCCs), integrated ingress routing, and pre-configured observability stacks.

OPENSHIFT WEB CONSOLE OVERVIEW



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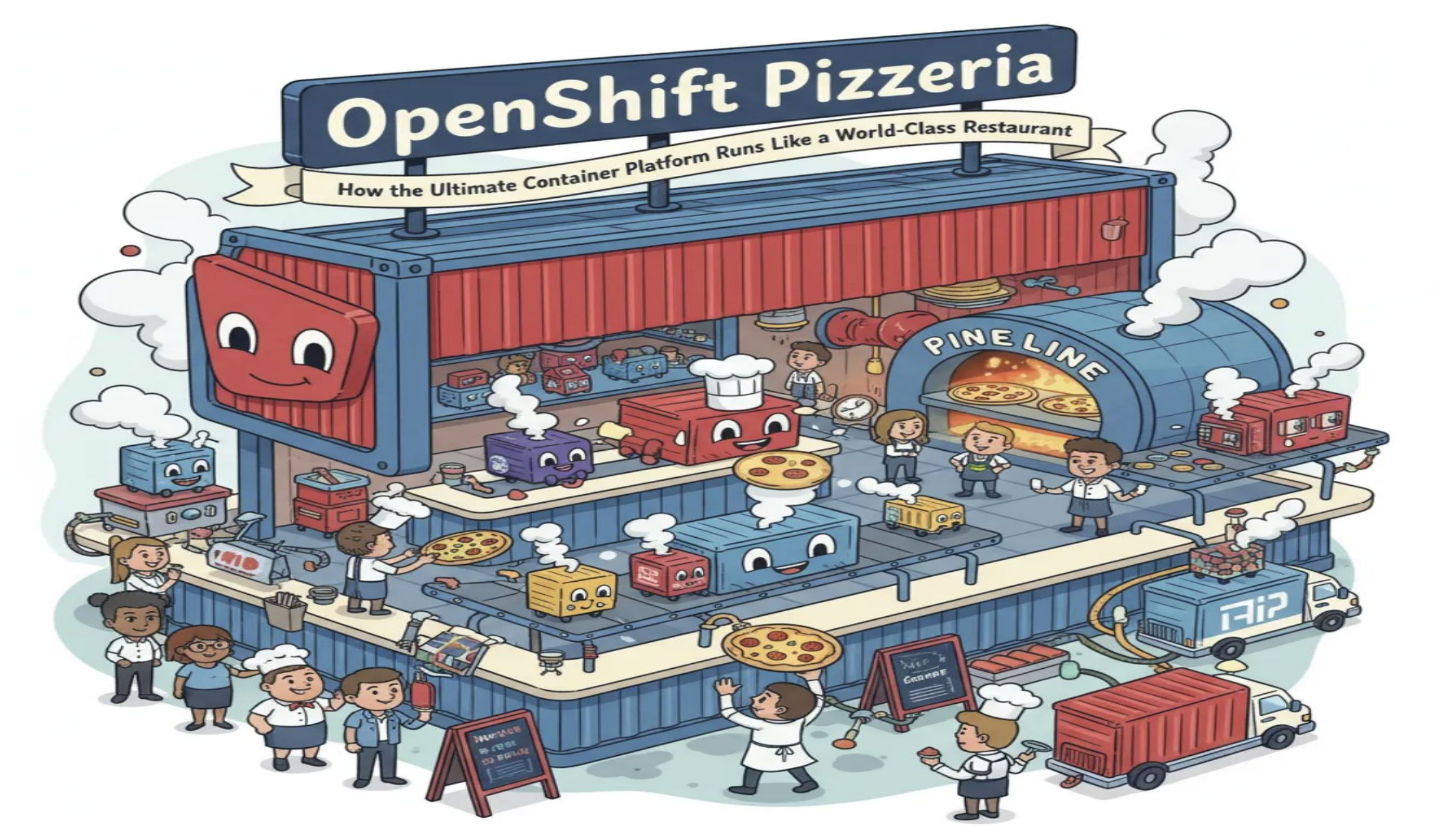

https://medium.com/@katekeiroz/openshift-pizzeria-how-the-ultimate-container-platform-runs-like-a-world-class-restaurant-ce1bd9fe4174?sharedUserId=katekeiroz

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Activity: "Build Your OpenShift Pizzeria"

Round 1 — "Design Your Kitchen" (15 min)

"You've been hired to open a new OpenShift Pizzeria branch. Your menu has **3 pizzas** (applications): **an ordering website, a payment API, and a delivery tracker**. Design your kitchen."

Round 2 — "The Dinner Rush" (10 min)

"Teams have 90 seconds per card to discuss and answer: **what happens in your pizzeria, and what OpenShift concept handles it?**"